

Temporal as Execution Substrate for the Ledger v16

An addendum: five-engineer consensus, v1.0

TE-1 ... TE-5 (temporal-engineer $\times 5$), converged

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1 The Way

Temporal runs the two *untrusted* machines and nothing else. The Event Monitor and the Events Executor become durable, replayable orchestration; the Transaction Executor and the immutable log stay outside Temporal as the trusted single writer, fronted by one idempotent propose-to-door activity — the only write path. The ledger log is the sole book of record. Temporal history is a disposable execution cache, and the acceptance test is operational: wipe the Temporal cluster, and the entire workflow population re-derives from the ledger — UnitStatus re-read, watches re-armed, nothing lost.

One workflow per unit. Each unit runs one long-lived workflow, keyed by its `unitId`, walking the unit's product graph. The workflow's pending timers and signal-waits equal the current node's out-edge set — the armed watch list of the graph-consistency invariant. The in-memory node is a mirror, never authoritative; it is rehydrated from the ledger on start, on `ContinueAsNew`, and on any doubt. Units the ledger itself creates — the settlement-obligation unit, a market-claim unit, the partial-split successor legs — start their own independent unit-workflows (signal-with-start, no Temporal parent-child coupling: parental linkage is ledger-absent relational state). Branches of a unit's own graph, such as the market-claim branch at a node, stay inside that unit's single workflow.

Side effects live in activities only. Capture an arrival, fetch a projection, evaluate a contract, propose to the door, emit a settlement instruction — each is an activity. Workflow code holds timers, signal-waits, selectors, and sequencing, and nothing else. Smart contracts run in version-pinned, queue-routed activities (never local activities), so ledger economics never enters Temporal's replay or versioning surface: the ledger replays by re-reading admitted transactions, Temporal by re-reading its own history, and the two never meet.

Watches. Date watches are in-workflow durable timers; observation and barrier watches are signal-driven evaluation against recorded observations, never a timer poll. Temporal Schedules serve system cadence only (end-of-day sweeps, overdue-watch reconciliation) — a 252-fixing swap is one workflow, not 252 schedules. The lifecycle declared→armed→fired/expired is authoritative in the ledger: arming writes the recorded acknowledgement through the door, and Temporal supplies only the liveness half.

The boundary. Capture precedes routing: a pushed arrival is recorded through the door — capture envelope, quarantine-or-route — by the ingestion activity before any workflow signal exists. Signals carry timing and reference (a cause identifier, a log position), never economic payload, and every signal has a record-derived backstop: a lost signal costs latency, never the record. The settlement interface remains a pure projection with no write privilege; settlement failure returns as a recorded external event and walks the settlement-obligation unit's graph — never a Temporal-side compensation or saga rollback.

Idempotence. The door’s cause-derived transaction identifier, computed from recorded inputs and never from Temporal run or attempt identifiers, absorbs Temporal’s at-least-once activity delivery into exactly-once commitment. Temporal’s own deduplication is tolerated as optimisation, never trusted for correctness.

2 Decision Register

Unanimous after three rounds. R-01..R-21 as amended in Round 2; R-22..R-27 added from Round-1 missing-decision returns.

- R-1.** Temporal runs only the two untrusted machines; the Transaction Executor is an external single-writer service fronted by one idempotent door activity (M6).
- R-2.** The ledger log is the sole book of record; Temporal history is disposable; wipe-and-rebuild from the ledger is the standing acceptance test (C-2.2).
- R-3.** One long-lived workflow per unit, keyed by plain `unitId` in the authoritative production namespace; lineage is a namespace-level fact, and `lineageId:unitId` is reserved for shared simulation namespaces where non-authoritative lineages deliberately co-locate.
- R-4.** The workflow’s node and watch set mirror the ledger and are rehydrated from it; pending timers and signal-waits equal the current node’s out-edge set.
- R-5.** Every side effect is an activity; workflow code is orchestration only.
- R-6.** Contracts run in version-pinned activities, never workflow code.
- R-7.** Idempotence is the door’s cause-derived txid from recorded inputs; at-least-once delivery becomes exactly-once at the door; Temporal dedup is never load-bearing.
- R-8.** Date watches are durable timers; condition watches are signal-driven evaluation over recorded observations, never polls.
- R-9.** Schedules serve system cadence only; per-unit contractual dates never use them.
- R-10.** The watch lifecycle is authoritative in the ledger; arming is a recorded acknowledgement through the door; Temporal supplies liveness (M1, M3).
- R-11.** Signals carry timing and reference only, never payload (M4), with record-derived backstops.
- R-12.** Capture precedes routing: envelope-first through the door before any signal (M8, TA-ARRIVAL); a signal is never the sole record of an arrival.
- R-13.** Updates serve human gates only (forward repair, quarantine disposition, authorised fork), returning the door’s admit/refuse; boundary capture never rides an Update; Queries are a non-authoritative liveness view.
- R-14.** Ledger-created units start their own independent unit-workflows (signal-with-start, no parent-child coupling); graph branches stay in the owning unit’s workflow; corporate-action cascades fan out as signals to existing units, each firing proposing its own txid. *Round-3 note: TE-2’s dispute withdrawn on the certified partial-edge text — the split legs are genuinely new units.*
- R-15.** ContinueAsNew at node transitions and every K fixings, carrying only {unitId, nodeId, log-cursor}; rehydration and disaster recovery are the same re-derivation.
- R-16.** No run/execution timeouts on unit workflows; liveness is bounded by watch windows and obligation deadlines surfaced as overdue items. Door activity: short start-to-close,

unlimited retry for transient failures only (see R-22). Long waits are timer-plus-signal; only egress and batch activities heartbeat.

- R-17.** Two versioning axes stay separate: Temporal Build-IDs/GetVersion for orchestration (ContinueAsNew boundaries as cutover points); ProductTerms versions on the log for economics, which alone bear C-2.2.
- R-18.** Task queues split by trust and scaling: door (sole write credential, serialized, backpressure-alarmed), contracts (read-only, versioned), ingestion (must-not-lose), unit-orchestration (idle-many), settlement (egress-only).
- R-19.** Search attributes (unit, node, next watch due, obligation status, overdue and quarantine flags) are an ops projection reconciled to the ledger, never authoritative.
- R-20.** Authorised forks and simulated ledgers run in isolated namespaces against their own lineage's door, never the production door (C-12.5, C-2.8).
- R-21.** Economic errors are repaired forward through the door; a Temporal non-determinism bug degrades to a liveness incident, never wrong ledger state.
- R-22.** A door *refusal* is a returned value, not a retryable error: it breaks the retry loop and stands as a visible blocked/overdue item awaiting declared precedence or a corrected proposal — never retried to infinity, never a crash.
- R-23.** A late or back-dated observation is an ordinary capture-through-the-door arrival; affected workflows re-fire on the correction signal under a new txid; the valid-time re-derivation is a pure ledger fold — Temporal never rewinds workflow state.
- R-24.** Settlement failure is a recorded external event walking the unit's graph, never a Temporal compensation; the settlement projection proposes nothing.
- R-25.** Signal-arrival order is orchestration sequencing only; the committed order is decided at the door alone (C-2.7).
- R-26.** EvaluateContract is a normal queue-routed activity, never a local activity.
- R-27.** Fan-out to possibly-not-running workflows uses signal-with-start; record-first capture remains the correctness backstop.

3 What Temporal Must Never Own

The book of record — the immutable, hash-chained log and every fact folded from it. Balances, the three homes, PnL, NAV, every projection. Invariant enforcement: conservation, coverage and its right-of-use gate, consistency of reference, the admission decision itself. The total order of transactions. Idempotence keys. The watch lifecycle's recorded facts (declared, armed, fired, expired). Quarantined events and capture envelopes. The close-out net and CLM-CO. Anything the ledger must reconstruct alone: if losing Temporal would lose it, the design is wrong — Temporal owns schedules-in-flight and orchestration position, and nothing else.

4 Operational Constraints

Determinism rules for workflow code. No wall clock (`workflow.Now` only, and never feeding a contract), no randomness, no iteration over unordered maps, no direct I/O, no local activities for contract evaluation; all non-determinism lives in activities whose results Temporal records.

Versioning plan. Orchestration changes ride Worker Versioning; old runs drain to their next ContinueAsNew boundary, which is the standing cutover point, so GetVersion patches are rare. Economics changes are new ProductTerms versions plus new version-pinned pure functions on the record; they never touch workflow code.

Timeouts. Door activity: start-to-close of seconds, unlimited retry on transient failure, refusal returned as a value (R-22). Contract evaluation: start-to-close bounded by pricing cost, heartbeat none. Ingestion capture: schedule-to-close tight, since capture loss is the one unacceptable latency. Egress and batch reconciliation: heartbeated. No workflow run or execution timeouts (R-16).

ContinueAsNew policy. At every product-graph node transition and every K fixings (K sized to keep history well under the server limit); carried state is the identifier triple only; everything else rehydrates from the ledger.

Worker topology. Five queue families (door, contracts, ingestion, unit-orchestration, settlement) with separate worker pools, credentials, and scaling; the door pool alone holds ledger-write credentials, sized to the single writer, with schedule-to-start latency alarmed as the backpressure signal.

5 Declaration

Rounds used. Round 0 (five independent papers), Round 1 (21-entry consolidated register: 18 unanimous, 3 disputed, 6 missing decisions raised), Round 2 (amended register: 49 of 50 rulings AGREE), Round 3 (single scoped fact-check on R-14 against the certified v16.0 partial-edge text; dispute withdrawn). **Final register status: UNANIMOUS on all 27 entries. No Decision Panel ruling was required.**

Decision Log. DL-01 | Phase 2 | authoritative v16 artifact | single candidate (`ledger_v16_0.tex`, certified) | no panel convened | n/a. DL-02 | Phase 2 | R-14 dispute (split legs: new units or branches?) | resolved by the certified specification text, not by vote; TE-2 withdrew | no panel convened | 5–0. DL-03 | Phase 3 | placement | repository convention (all cross-cutting documents live beside the Manifesto: TuringReview precedent) | not genuinely ambiguous; no panel convened | n/a. **DEFERRED-TO-OWNER: none** — the substrate proposal demands nothing of the Constitution; every entry keeps the ledger as sole truth.

Independence caveat (recorded, not absorbed). The five instances share the temporal-engineer agent type’s persistent memory; TE-5 disclosed consulting it. Round-0 independence is therefore partial. Mitigation: the adversarial round structure, the register-level dispute discipline, and the Round-3 fact-check against the specification text rather than any instance’s recollection.

Confidence: HIGH on the architecture mapping and the negative space (they restate the specification’s own trust boundary); **MEDIUM** on the operational constants (K for ContinueAsNew, timeout values), which are estimates pending a load model — the single biggest unknown is production event volume per unit per day, which sizes history growth, the door pool, and the ContinueAsNew cadence.